

tudes of about 160km and are used for Earth observation and surveillance.

Geostationary. These orbit directly above the Equator with a period equal to one Earth day, and as a result appear to be fixed. These are widely used for TV communication.

Geosynchronous. These are very similar to geostationary ones except that their orbit is centred over a fixed latitude and is in the shape of number eight (8).

Polar. These orbit Earth from

to identify common stars that are used as reference for navigation, magnetometer, GPS, accelerometer, gyroscope and horizon sensors. All these sensor inputs are analysed by the control system to stabilise and keep the satellite on the designated path, counteracting external torque and forces.

Telecommunication. This sub-system is used for RF communication either with the base station or between satellites. It contains a

is converted to various other voltage rails as per the demand from each of the sub-systems.

Onboard computer and data handling. This is the brain of the satellite. It has a processor, RAM and associated memories that enable it to perform a variety of functions. It acquires data from the payload and processes it based on the running algorithm.

It can then store this data on-board and/or relay it to the ground station. Along with this, it aids ADCS sub-system to process sensor inputs to decide the correct course of action. It also monitors other related sub-systems and keeps a track of the running processes. Usually, the sub-systems are connected on a standard 1553 or low-voltage differential signaling interface.

Actuator and propulsion electronics. This is an electromechanical system that implements the decision taken by ADCS sub-system. It usually fires the booster stage or enables the motor drives that bring about rotation of the resonant wheels, which adjust the course of the satellite.

Payload. This is the sole purpose of the flight mission. The type of payload determines the type of satellite. Humans can also be part of the payload. A few common types of payload include meteorology, GPS system, satellite based telecommunication and synthetic aperture radar used to map a planet's surface with great degree of accuracy. Fig. 2 illustrates a full-blown satellite along with its launch vehicle.

The term satellite might sound intimidating at the beginning but we hope that the article has made it a better discernible concept using the sub-systems approach. Each sub-system described here has a number of smaller systems within it. Harmonic interoperation of these systems ensures success of the mission, which is ultimately another step of man towards exploring eternal Space. **EFY**

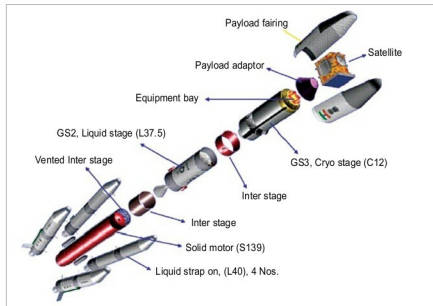


Fig. 2: Exploded view of a satellite

pole-to-pole, providing meteorological information.

A satellite is launched for a particular mission, which makes up the payload of the satellite. This payload is carried into the orbit as a part of the satellite by the launch vehicle. If we consider a typical satellite, it would contain the blocks shown in Fig. 1. Each of the sub-systems has its own significance and is an indispensable part of the satellite.

Attitude determination and control system

This is the navigation and stabilisation system of the satellite. Attitude determination requires multiple sensors like Sun sensor to determine satellite's position with respect to Sun, star sensors

modulator block to transmit data and a demodulator to receive and process the data. This block is often used to control the satellite from the ground station and to transmit/receive data to/from the satellite.

Electronic power system. The satellite has a lot of electronic content in it, which needs to be powered over its lifetime of operation. This sub-system mainly contains two sections: generation and battery charging, and distribution and monitoring.

The sub-system uses solar panels to generate electricity from Sun and stores it in batteries. A monitored power bus from the battery is distributed across other sub-systems. This voltage is usually 28V, 42V or 70V. It